

Topic 3

Coordinate geometry

Concept guide · Worked examples · Cambridge-style practice · Fully explained answers

Learning focus

Explain every method before practising. Use the original practice questions to check calculation, reasoning and communication.

Topic map

3.1	Coordinates and gradients
3.2	Distance and midpoint
3.3	Equations of straight lines
3.4	Parallel and perpendicular lines
3.5	Intersection and geometric reasoning

Core foundations and Extended applications are integrated throughout. Study each example, then attempt the practice paper without the notes.

1. Gradient from two points

The gradient of the line from (x_1, y_1) to (x_2, y_2) is change in y divided by change in x . A positive gradient rises to the right and a negative gradient falls. A horizontal line has gradient zero; a vertical line has undefined gradient.

Worked example

From $(2, 3)$ to $(6, 11)$, $m = (11 - 3)/(6 - 2) = 2$.

Exam check: Keep coordinate subtraction in the same order in numerator and denominator.

2. Midpoint and distance

The midpoint averages the two x -values and the two y -values separately. The distance formula comes from Pythagoras: $d^2 = (\Delta x)^2 + (\Delta y)^2$. A sketch helps detect an impossible negative distance or midpoint outside the segment.

Worked example

From $(1, 2)$ to $(7, 10)$, midpoint = $(4, 6)$ and distance = $\sqrt{6^2 + 8^2} = 10$.

Exam check: Distance is non-negative, even if the line has a negative gradient.

3. Equation of a line

Use $y = mx + c$ when the gradient and vertical intercept are known. If a point (x_1, y_1) and m are given, substitute into $y - y_1 = m(x - x_1)$, then simplify. To find x -intercept set $y = 0$; to find y -intercept set $x = 0$.

Worked example

Gradient 3 through $(2, 7)$: $y - 7 = 3(x - 2)$, hence $y = 3x + 1$.

Exam check: Substitute the original point to check the final equation.

4. Parallel and perpendicular lines

Parallel non-vertical lines have equal gradients. Perpendicular non-vertical lines have gradients whose product is -1 . Horizontal and vertical lines are perpendicular to each other, even though the vertical gradient is undefined.

Worked example

A line perpendicular to $y = 2x + 1$ has gradient $-1/2$. Through $(0, 3)$ its equation is $y = -x/2 + 3$.

Exam check: Taking the negative alone is not enough; take the negative reciprocal.

5. Intersection and coordinate proofs

The intersection of two lines satisfies both equations. Solve simultaneously, then substitute the result into both. To prove a quadrilateral is a parallelogram, compare gradients of opposite sides or find matching midpoints of its diagonals. To check right angles, multiply gradients.

Worked example

For $y = 2x + 1$ and $y = -x + 7$, $2x + 1 = -x + 7$ gives $x = 2$; $y = 5$.

Exam check: An accurately drawn graph supports a result, but use algebra if exact coordinates are requested.

Before the practice paper

Method and accuracy

Write the formula or first operation, show the intermediate step, give units when needed and check by substitution, estimation or a sketch. On non-calculator work, keep exact forms when requested.

Practice paper

Topic 3 · Coordinate geometry

Instructions

Original Cambridge-style practice. Answer all questions. Section A is multiple choice; Section B requires working and final answers. Try Section A without a calculator; use a calculator in Section B only where needed.

Candidate name:	
Class / date:	

Total: 31 marks

Section A · Multiple choice

Choose one answer for each question.

1. Gradient from (0,1) to (2,5)? [1]

- A 1
- B 2
- C 3
- D 4

2. Midpoint of (2,4) and (8,10)? [1]

- A (6,14)
- B (5,7)
- C (3,3)
- D (10,14)

3. Distance between (0,0) and (3,4)? [1]

- A 4
- B 5
- C 6
- D 7

4. Which line is parallel to $y = 3x + 2$? [1]

- A $y = -3x + 4$
- B $y = x/3 + 4$
- C $y = 3x - 7$
- D $y = -x/3 + 1$

5. Which gradient is perpendicular to -4? [1]

- A 4
- B $-1/4$
- C $1/4$
- D -4

6. What is the y-intercept of $y = -2x + 9$? [1]

- A -2
- B 2
- C -9
- D 9

7. For $y = 2x - 6$, the x-intercept is [1]

- A -3
- B 0
- C 3
- D 6

8. Which is the equation of the horizontal line through (4,-2)? [1]

- A $x = 4$
- B $y = 4$
- C $x = -2$
- D $y = -2$

Section B · Structured questions

9. A(1,2) and B(7,10) lie on a straight line.

(a) Calculate the gradient of AB.

[2]

(b) Calculate the midpoint of AB.

[2]

(c) Calculate the length AB.

[2]

[Total: 6]

10. A line passes through (2,5) and has gradient -3.

(a) Find its equation in the form $y = mx + c$.

[3]

(b) State its y-intercept.

[1]

(c) Find its x-intercept.

[2]

[Total: 6]

11. The lines $y = 2x + 3$ and $y = -x + 12$ intersect.

(a) Find the x-coordinate of their intersection.

[2]

(b) Find the y-coordinate.

[1]

(c) Check the point satisfies both equations.

[2]

[Total: 5]

12. P(0,4), Q(6,7) and R(8,3) are three vertices of a quadrilateral.

(a) Find the gradient of PQ.

[2]

(b) Find the gradient of QR.

[2]

(c) Explain whether PQ is perpendicular to QR.

[2]

[Total: 6]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Award method marks when the mathematical setup is correct even if an arithmetic slip follows.
Check each part independently.

Section A · Answers and reasons

1. B — 2. Change in y is 4 over change in x of 2.
2. B — (5,7). Average corresponding coordinates: $(2+8)/2$ and $(4+10)/2$.
3. B — 5. Pythagoras gives $\sqrt{3^2+4^2} = 5$.
4. C — $y = 3x - 7$. Parallel lines have the same gradient of 3.
5. C — $1/4$. A perpendicular gradient satisfies $(-4)m = -1$.
6. D — 9. Set $x = 0$; the remaining y-value is 9.
7. C — 3. Set $y = 0$ and solve $2x = 6$.
8. D — $y = -2$. A horizontal line has a constant y-coordinate of -2.

Section B · Worked solutions

9. a) $(10 - 2)/(7 - 1) = 4/3$ [2]. b) $((1 + 7)/2, (2 + 10)/2) = (4, 6)$ [2]. c) $\sqrt{6^2 + 8^2} = 10$ [2].

10. a) $y - 5 = -3(x - 2)$, so $y = -3x + 11$ [3]. b) 11 [1]. c) $0 = -3x + 11$, $x = 11/3$ [2].

11. a) $2x + 3 = -x + 12$, $3x = 9$, $x = 3$ [2]. b) $y = 9$ [1]. c) $2(3) + 3 = 9$ and $-3 + 12 = 9$ [2].

12. a) $(7 - 4)/(6 - 0) = 1/2$ [2]. b) $(3 - 7)/(8 - 6) = -2$ [2]. c) $(1/2)(-2) = -1$, therefore perpendicular [2].

Review your errors

Mark the first wrong step: setup, algebra, arithmetic, units, graph or reasoning. Redo that part without seeing the solution.

Syllabus reference: Cambridge IGCSE Mathematics 0580, 2025-2027,
www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf. Paper style informed by the June 2026 papers and examiner report provided by the user. All practice questions are original.